PAUL ROBINSON

Draftsperson | Mechanical Engineer | Sheet Metal Fabricator

0421 847 001 | probinson85@live.com.au | Pimpama, QLD 4209 | linkedin.com/in/paulrobinson1985

PROFESSIONAL SUMMARY

A mechanical engineer with an unusual foundation: four years as an apprentice sheet metal fabricator and a further five years in the trade before completing a Bachelor of Mechanical Engineering with Honours at Griffith University in 2023. That combination — trade hands and engineering mind — means I design parts knowing exactly how they will be made. SolidWorks has been my primary CAD platform throughout my degree and subsequent design work, with particular experience in sheet metal and weldment design, complex multi-level assemblies, and designing adaptive devices for manufacture. Currently consulting at Habitec, creating custom engineering solutions from concept through to fabrication. Off-road vehicles, fabrication, and problem-solving in the field are a central part of my life outside work — I'm presently modifying an enclosed trailer into a fully custom camper and have spent years building and modifying 4WDs for serious off-road use.

KEY SKILLS

CAD & Design	SolidWorks (sheet metal, weldments, multi-level assemblies, 3D printing design) — primary platform throughout engineering degree and professional work
Additional CAD	Autodesk Inventor, Fusion 360 — proficient across multiple platforms, quick to adapt to new software
Fabrication	Sheet metal fabrication, MIG/TIG welding, stainless steel, mild steel and aluminium — 4-year apprenticeship + 5 years trade experience
Standards	AS and ISO welding and materials standards, engineering drawing standards — Griffith University BE programme
Documentation	Technical report writing, engineering specifications, SOP development, supplier communication and first-off procurement coordination
Product Design	Novel product design from concept to manufacture — adaptive devices, expedition equipment, custom fabrications

WORK EXPERIENCE

Habitec — Mechanical Engineering Consultant *Jul 2020 — Present*

- Design and engineer custom adaptive devices for individuals with disabilities, working from identified user need through to final manufactured solution — no templates, every project designed from scratch.
- Use SolidWorks throughout the full design cycle: concept modelling, sheet metal and structural design, assembly validation, and producing manufacturing drawings for in-house or external fabrication.
- Coordinate with external suppliers for first-off prototype fabrication, managing quotes, purchase orders and technical communication to ensure manufactured parts match design intent.
- Produce detailed engineering documentation including specifications, design rationale reports, and technical handover packages for multidisciplinary teams.
- Conduct client assessments and translate real-world functional requirements into engineered solutions — working autonomously with minimal oversight while managing multiple concurrent projects.

Superior Stainless — Fabricator / Welder *Jun 2009 — Oct 2011*

- Led specialised fabrication projects in stainless steel, mild steel and aluminium, with a focus on medium to light sheet metal processing requiring precise dimensional control.

- Interpreted and problem-solved complex shop drawings, identifying manufacturing issues before cutting commenced — a skill now applied in reverse when producing drawings for others to build from.
- Contributed to team workflow and output quality across a busy fabrication department.

Swift Marine — Apprentice Sheet Metal Fabricator *Aug 2004 — Aug 2008*

- Completed a four-year Certificate III Sheet Metal Fabrication Apprenticeship, developing hands-on expertise across the full range of fabrication processes: layout, cutting, forming, welding, and finishing.
- Gained experience fabricating complex marine components requiring tight tolerances and corrosion-resistant materials — a strong foundation for weldment and sheet metal design work.

Acran | Metal Master Products — Fabricator / Rigger / Installer *Aug 2008 — Jun 2009*

- Noise control device installation (Acran) — sheet metal fabrication, rigging, brake press and guillotine operation, MIG welding in an industrial environment.
- Commercial fabrication at Metal Master Products: toolboxes, stainless steel water features, TIG and MIG welding across a range of products.

Prior Career — Crane Operations | 2010–2015

Prior to commencing my engineering degree, I had a five-year career as a crane operator across Queensland, South Australia and Western Australia — operating all-terrain fleets from 30T to 280T across major construction, infrastructure and mining projects. This career ended following a spinal cord injury in 2015, after which I enrolled at Griffith University. Employers included Big Lift Cranes, Select Plant Hire, NQ Group, AllState Cranes, East Coast Development Engineering, and Hamon.

EDUCATION

Bachelor of Mechanical Engineering (Honours)
Griffith University, Gold Coast | 2023
Thesis: Designed a leg/foot attachment enabling passive cycling for individuals with lower limb paralysis — including full SolidWorks modelling, biomechanical load analysis, adaptive hardware design, and prototype iteration.

Certificate III — Engineering Fabrication
TAFE Queensland, Southport | 2007

Introduction to CNC Machining
TAFE Queensland, Southport | 2023

AWARDS & RECOGNITION

Griffith College Academic Excellence Award | 2017

CTQ Excellence Despite Adversity Award | 2016

REFERENCES

Soo Oh
Habitec — Project Manager
0401 390 359

Jim Barrett
Making Strides SCI Recovery Centre — Director
0401 331 730

Kyle Mulholland
BioSpine Project — Project Manager
0412 366 028

Adam Evans
East Coast Cranes — Operations Manager
0437 444 33